CS3 Rubric – Charlottesville Crime & Temperature Case Study

DS 4002 – Instructor: Dean Paler
Due: December 9

Submission format: GitHub repository link submitted on Canvas

Individual Assignment

Why am I doing this? Local community leaders often speculate about the relationship between public safety and weather. This case study provides you the chance to utilize your technical and analytical skills to investigate whether hotter days are associated with higher crime counts in the Charlottesville area. By applying correlation analysis and visualization, you will help answer a pressing local question: does temperature correlate with crime?

What am I going to do? Using the case study repository and template script, you will begin by cleaning and familiarizing yourself with two datasets: five years of daily crime reports and daily weather data from the Meteostat Python library. From there, you will aggregate crime counts by date, merge them with temperature records, and finally calculate the Pearson correlation coefficients to test the strength of temperature and crime's relationship. Along the way, you will also produce visuals. Deliverables include:

- A final GitHub repository including an updated Python script and slide deck that communicates your analysis to both city leaders and fellow students

Tips for success:

- Follow your curiosity! Explore the questions that arise as you examine the sources and data.
- Try new approaches! The template script is only a guide; feel free to experiment with different analyses or visualizations.
- Be transparent! Clearly document your assumptions, limitations, and any potential biases.

How will I know I have succeeded? You will meet expectations on this case study when you follow the criteria in the rubric below.

Spec Category	Spec Details
Formatting	- Repository – a GitHub repo containing all modified or produced materials - README.md - LICENSE.md - Organize your repository using the following folders: - Data – raw data exactly as provided in the case study repo - Scripts – the updated/modified Python notebook from the case study repo - Output – all graphs/visuals and the slide deck summarizing your findings
README.md	<u>Goal:</u> This file should orient anyone who visits your repository and

<table>
<tr><td></td><td>help readers understand the case and navigate the contents

Use Markdown headers and include the following H2 sections:

Software and platform
List all software, programming languages, and add-on packages used, including those already included in the template Python notebook
Documentation map
Provide a clear outline or directory tree showing the hierarchy of folders and subfolders in your repository; list all files contained in each section
Reproduction instructions
Give step-by-step instructions that allow someone else to fully reproduce the results
References
Include citations or links to any referenced materials, datasets, or external sources; use IEEE Documentation style

</td></tr>
<tr><td>LICENSE.md</td><td>
<u>Goal:</u> This file explains to a visitor the terms under which they may use and cite your repository
Select the MIT license from the GitHub options list on creation
</td></tr>
<tr><td>DATA folder</td><td>
<u>Goal:</u> This folder contains all the data used for this project
Include the raw data from the case study repo
</td></tr>
<tr><td>SCRIPTS folder</td><td>
<u>Goal:</u> This folder contains all scripts used to transform the raw data into the final outputs (your updated/modified Python notebook)
Use the provided Python notebook as a starting point to clean the source data, conduct the Pearson correlation analysis, and generate the visualizations, resulting in the following output:

Two exploratory graphs
Pearson correlation coefficient between temperature and overall crime
Pearson correlation coefficient between temperature and specific offenses

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<tr><td>OUTPUT folder</td><td>
<u>Goal:</u> This folder contains the slide deck and all output produced by the case study and communicates your findings

Create a slide deck presentation that guides an uninformed audience through your work including

Title and outline
Context and research question
Data explanation
Modeling approach and analysis plan
Limitations
Results and conclusions, with any tables and figures

Numeric outputs from updated script
Interpreted Pearson correlation

Next steps
References
Closing slide

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Acknowledgements: Special thanks to Jess Taggart from UVA CTE for coaching on making this rubric.
This structure is pulled from Streifer & Palmer (2020).